CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 6: **We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a square of side $(x + 4)$ is: 1

(a) $x^2 + 4x + 16$ **(b)** $x^2 + 16$ **(c)** $4x + 16$ **(d)** $x^2 + 8x + 16$

2. $(x + 6)^2$ equals: 1

(a) $x^2 + 12x$ **(b)** $x^2 + 6x + 36$ **(c)** $x^2 + 36$ **(d)** $x^2 + 12x + 36$

3. $(x - 9)^2$ equals: 1

(a) $x^2 - 9x + 81$ **(b)** $x^2 - 81$ **(c)** $x^2 + 18x + 81$ **(d)** $x^2 - 18x + 81$

4. $(x + 11)(x - 11)$ equals: 1

(a) $x^2 + 22x - 121$ **(b)** $x^2 - 121$ **(c)** $x^2 + 121$ **(d)** $x^2 - 22x + 121$

5. $(x + 1)(x + 8)$ equals: 1

(a) $x^2 + 9x + 9$ **(b)** $x^2 + 8x + 9$ **(c)** $x^2 + 8$ **(d)** $x^2 + 9x + 8$

6. 104×96 equals: 1

(a) 9984 **(b)** 9992 **(c)** 10016 **(d)** 9996

7. In the expansion of $(2x + 7)(3x + 2)$ the coefficient of x is: 1

(a) 20 **(b)** 25 **(c)** 21 **(d)** 4

8. The product of two numbers $(a + b)$ and $(a - b)$ is: 1

(a) the sum of the squares of a and b

(b) the difference of the squares of a and b

(c) the square of $a - b$

(d) twice the product of a and b

9. $(x + 7)^2$ equals: 1

(a) $x^2 + 7x + 49$ **(b)** $x^2 + 14x$ **(c)** $x^2 + 14x + 49$ **(d)** $x^2 + 49$

10. $(x - 10)^2$ equals: 1

(a) $x^2 - 10x + 100$ **(b)** $x^2 - 20x + 100$ **(c)** $x^2 + 20x + 100$ **(d)** $x^2 - 100$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find 45×55 using $(a + b)(a - b)$. 2

12. Expand $(x + 12)^2$ using an identity. 2

13. Expand $(5x + 2)(3x + 2)$. 2

14. Find 86×94 using $(a + b)(a - b)$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Expand $(2x + 7)^2$. 3

16. Find 74×86 using the identity $(a + b)(a - b) = a^2 - b^2$. 3

17. Simplify $(x + 2)(x + 8) + (x - 2)(x - 8)$. 3

OR

Expand $(3x + 2)^2$. 3

18. Find 118×122 using the identity $(a + b)(a - b) = a^2 - b^2$. 3

OR

Simplify $(x + 5)(x + 4) + (x - 5)(x - 4)$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangle has length $(x + 5)$ cm and breadth $(x + 8)$ cm. Write an expression for its area and find the area when $x = 2$. 5

OR

Prove that $(a + b)^2 - (a - b)^2 = 4ab$ and verify it for $a = 10$, $b = 3$. 5

20. A rectangle has length $(x + 3)$ cm and breadth $(x + 4)$ cm. Write an expression for its area and find the area when $x = 10$. 5

OR

Prove that $(a + b)^2 - (a - b)^2 = 4ab$ and verify it for $a = 12$, $b = 5$. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND